

# MedFinder – Real-Time Medicine Availability & Store Locator

Project Proposal for UCS503P

Submitted to: Mr. Asif

**Thapar Institute of Engineering and Technology**

Project Team

Sourabh Chhabra(1024030435) | Madhav Gupta(1024030436) | Arpit Yadav (1024030446)

August 16, 2026

## 1. Higher-order goal

*...secondary framing*

This project aims to improve access to essential healthcare by reducing the time patients spend searching for medicines during urgent or routine needs. While the underlying issue of pharmacy inventory visibility spans an entire city, the immediate engineering objective is to translate real-time stock and location transparency into a measurable, deployable software solution.

## 2. Time-to-value

*...primary heuristic*

- We will deliver meaningful increments quickly by prototyping the core search-and-locate workflow and validating it with users within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration — a working medicine search and store locator — then improved based on user feedback.

## 3. Problem Statement

Patients in urban and semi-urban areas frequently struggle to find medicines quickly, especially during urgent medical needs. Common pain points include:

- Fragmented availability: not every pharmacy stocks every medicine, and there is no way to know which store has a specific item without visiting or calling around.
- Wasted time and effort: patients often roam from store to store, which delays treatment and worsens outcomes, particularly for the elderly or critically ill.
- Unexpected stockouts: even common, everyday medicines go out of stock without warning, compounding the same problem.
- No visibility before travel: there is no simple, centralized way to check medicine availability across nearby stores before stepping out of the house.

Why this matters now (contextual relevance): In cities where pharmacies operate independently with no shared inventory system, even a lightweight visibility layer can meaningfully cut down search time and reduce delays in receiving treatment.

## 4. Proposed Solution

*...Software Proposition*

### 4.1 Overview

Build a web-based “Search-to-Pickup and Delivery” platform with the following components:

- Store & inventory listing: recognized medical stores register and maintain a live inventory of the medicines they stock.
- Medicine search: users search for a medicine and see which nearby stores have it in stock, along with price and distance.
- Filtering & sorting: results can be filtered/sorted by nearest distance and lowest price.
- Reserve feature: users can place a short-hold reservation on a medicine at a chosen store so it is not sold out before they arrive.
- Verified registration: only recognized, verified pharmacies can list inventory, keeping the platform trustworthy.

### 4.2 Core workflow

- User searches for a medicine and instantly sees a ranked list of nearby stores with stock, price, and distance.
- System checks store-submitted inventory data and flags low-stock or recently-updated listings.
- User optionally reserves the medicine at a chosen store, receiving a reservation ID and hold window.
- Store staff confirm the reservation, update stock upon pickup or expiry, and the record closes out.

### 4.3 Operational constraints for an educational setting

- Keep the solution usable on low-to-mid range devices via responsive web design.
- Avoid dependence on advanced research algorithms (e.g., no ML-based demand forecasting in v1); focus on workflow, data correctness, and search performance.
- Design for deployability using common web hosting and managed databases.

## 5. Solution Approach

*...Engineering focus*

This is an engineering course project, so we focus on scalable data-management and workflow aspects rather than novel algorithm research.

### 5.1 Search & matching

*...non-research*

- Medicine search will use straightforward, well-understood techniques rather than novel research: fuzzy text matching on medicine name/brand/generic name to tolerate typos and partial queries.
- Distance ranking will use standard geolocation math (haversine distance) against store coordinates, without claiming state-of-the-art routing.

- Inventory freshness will be tracked via simple “last updated” timestamps per store-medicine record, surfaced to users so stale listings are visible rather than hidden.

## 5.2 Web architecture

*...web-based solution*

A typical 3-tier web architecture:

- Frontend: responsive UI for patients (search, filters, reservation) and store staff (inventory management dashboard).
- Backend API: authentication, medicine search, inventory CRUD, reservation logic, store verification workflow.
- Database: store users, stores, medicines, inventory records, reservations, and price history.

## 5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features (e.g., adding the Reserve feature after core search is stable).

## 6. Evaluation Criterion

*...measurable and attributable*

We define success using measurable metrics tied to the core workflow.

### 6.1 Primary evaluation metric

Time-to-find (TTF): time between a user submitting a medicine search query and the app returning a usable list of in-stock nearby stores.

- Target: median TTF  $\leq 3$  seconds for search results during pilot window.
- Attribution: measured via server-side request/response timestamps logged for each search query.

### 6.2 Secondary evaluation metrics

- Reservation fulfillment rate: percentage of reservations successfully honored (medicine still available) at pickup versus expired/cancelled.
- Search success rate: percentage of searches that return at least one in-stock result, compared against a pre-launch baseline (phone-call survey of store availability).
- User clarity: patient-reported clarity/usefulness of results using a simple 1–5 feedback question.
- Reliability: availability of staging/production for search and reservation during business hours (e.g.,  $\geq 99\%$  uptime in pilot).

### 6.3 Pilot validation plan

*...high-level*

- Recruit 10–20 pilot users (patients) and 3–5 partner pharmacies (or simulated store accounts if real partners are unavailable).
- Compare metrics before/after within the iteration window by logging baseline estimates via short interviews or manual store-calling exercises.

## 7. Scalability

*...with foresight*

The proposition should scale in both theoretical and practical senses.

### Scalable (theoretically)

The system should support growth in number of stores, medicines, and concurrent searches by:

- decoupling components (frontend/backend),
- enabling pagination and indexing for common queries (e.g., medicine name, store location),
- using stateless API design.

### Operationally deployable (practically)

The system should run on common web hosting:

- managed database,
- containerized or platform-hosted deployment,
- CI/CD pipeline for staging/production.

## 8. Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a web-based project:

- Web framework for REST APIs and reactive frontend pages.
- Managed relational or document database.
- Authentication approach (token-based or session-based) with separate roles for patients and store staff.
- Geolocation/maps service for distance calculation and map display.
- Test framework for unit/integration tests.
- CI runner and staging deployment target.

We will use these mature components to focus on workflow design, correctness, and measurement rather than inventing new infrastructure.

## 9. Project scope and deliverables

*...iteration-friendly*

### 9.1 Initial deliverable

*...first iteration*

- Store registration and inventory submission form
- Medicine search with distance and price filters
- Patient results page: store list with stock status, price, distance
- Basic logging and timestamps for TTF measurement
- CI: build + backend unit tests + linting
- CD to staging with a single command or pipeline job

### 9.2 Subsequent deliverables

- Reserve feature with hold-timer and expiry handling
- Notification layer (SMS/email optional in later iteration; can start with in-app notifications)
- Store dashboard for inventory management and reservation metrics
- Improved search/filtering and indexed queries for scale readiness

## **10. Risks and mitigations**

- Data accuracy: store-submitted inventory may go stale; mitigate with “last updated” timestamps and periodic reminder prompts to stores.
- Store adoption: provide an easy UI for inventory updates; keep the update workflow to a few taps; consider bulk-upload/CSV import.
- Verification overhead: manual pharmacy verification could bottleneck onboarding; mitigate with a lightweight document-upload + admin-approval flow.
- Evaluation measurement ambiguity: instrument timestamps and define clear search/reservation states before development begins.
- Operational issues in pilot: use staging early; ensure CI/CD produces repeatable deployments.

## **11. Summary**

This project proposes a deployable web platform that gives patients real-time visibility into medicine availability across nearby pharmacies. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially time-to-find). It remains focused on engineering workflow, data correctness, and scalability readiness rather than research-driven novelty.